Juneha Hwang

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Education

Korea Advanced Institute of Science and Technology (KAIST) August 2023-
Bachelor of Science in Computer Science (GPA: 4.01 / 4.30) *Daejeon, South Korea*

Experience

Visual AI Lab (KAIST) July 2026–Sep 2026
Internship *Daejeon, South Korea*

- Designed and built multimodal AI agent that answers spatial questions about images/videos through Python execution, integrating vision-language models with object segmentation, tracking, and 3D reconstruction.
- Built reproducible evaluation infrastructure supporting 10+ benchmarks. Used component ablations, human-assisted verification, and execution-trace analysis to identify evidence-selection and error-recovery failures for better agent design.

Linq Alpha Jan 2025–Feb 2026
Data Engineer *Seoul, South Korea*

- Built and maintained private financial-data RAG pipelines for institutional investors, ingesting and structuring earnings calls, broker research, and investment notes for downstream retrieval and LLM workflows.
- Built a parallel-prefix-style algorithm for long-document metadata extraction, propagating preceding context concurrently, reducing latency by around 80% while preserving context-dependent LLM processing.
- Designed a multi-stage entity-resolution system mapping ambiguous multilingual company references to canonical FIGIs, combining lexical/vector retrieval with ticker, alias, and document-context for disambiguation.
- Productionized long-running LLM and data workflows with FastAPI, Celery/Redis, and PostgreSQL, adding asynchronous processing, schema migrations, and observability for reliable operation.

Projects

DEX-DEX Perpetual Futures Arbitrage Bot | *C++20*, *Boost.Asio/Beast*, *WebSockets*, *OpenSSL*

- Built a C++20 perpetual-futures arbitrage engine spanning 60+ instruments across decentralized exchanges, integrating real-time WebSocket feeds, opportunity detection, and concurrent execution with fill recovery.
- Eliminated per-message heap allocations using reusable buffers and selective JSON decoding, reducing p99 processing latency from 2.75 to 0.58 μs and increasing single-threaded throughput $5\times$.
- Implemented bounded single-producer/single-consumer logging with acquire/release synchronization and cache-line-separated cursors, sustaining 100K records/second at under 3 μs p99 latency in real runs.

Papers

FinAgentBench: A Benchmark Dataset for Agentic Retrieval in Financial Question Answering

- C. Choi, J. Kwon, A. Lopez-Lira, C. Kim, M. Kim, **J. Hwang**, et al.
- Published at **ACM ICAIF 2025**, Singapore, Singapore

Structuring the Unstructured: A Multi-Agent System for Extracting and Querying Financial KPIs and Guidance

- C. Choi, A. Lopez-Lira, Y. Lee, J. Kwon, M. Kim, **J. Hwang**, et al.
- Published at **FinIR Workshop @ ACM SIGIR 2025**, Padua, Italy

Honors

Competitive Programming

- Samsung Collegiate Programming Cup (SCPC) 2024, 2025, 2026 Finalist
- International Collegiate Programming Contest (ICPC) Seoul Regional 2025 Finalist, Taichung Regional 2025 Finalist (Silver Medal)

Other Honors

- Jane Street ETC 2024 1st Place
- Fall 2023 Dean's List (*KAIST*)

Technical Skills

Programming Languages: Python, C / C++, Rust, Scala, JavaScript, TypeScript, SQL, HTML/CSS
Tech Stack: PyTorch, Scikit-Learn, NumPy, Pandas, LangChain, LangGraph, FastAPI, PostgreSQL, DuckDB, Celery, Redis, Docker, Kubernetes, Git
Languages: Korean (native), English (native), Japanese (conversational)